

Intellectual Property Rights (IPR) Document: SUTRAVA

1. Introduction-

The SUTRAVA framework is an AI-powered platform for automated requirements engineering and is designed to minimize the human efforts required in software requirement elicitation. The framework helps overcome many limitations of the conventional approach in software requirements elicitation, such as ambiguity, time wastage, and poor communication between stakeholders and development teams.

The SUTRAVA framework uses sophisticated NLP and machine learning models to transform the unstructured communication by the stakeholders into structured and precise software requirements. This process not only facilitates better collaboration between the stakeholders and developers but also makes the process of requirement elicitation more efficient and effective. [3,5,6]

2. What SUTRAVA Does?

SUTRAVA accepts varied formats of unstructured data from stakeholders' inputs available from project management platforms like Atlassian Jira and Slack and transforms them into structured agile user stories. The SUTRAVA framework operates on an eight-step workflow methodology for capturing and refining stakeholder requirements. The result is well-defined and actionable user stories that can be used in the entire software development process.

1. Ingestion

2. Segmentation

3. Detection

4. Named Entity Recognition (NER)

5. Structuring

6. Clustering

7. Prioritization

8. Explainability

Entities, such as stakeholders, system components, and constraints on the project, are automatically recognized by the framework using the obtained requirements. The framework categorizes the requirements that are semantically similar to each other in order to reduce redundancies. Each requirement is given a priority, and an explanation regarding why it has been prioritized is also provided.

3. Novelty and AI Innovations?

One of the key differences between SUTRAVA and other similar tools is the AI architecture, which was developed solely for the purposes of RE and not outsourced from third-party LLMs. Such an approach allows for more accurate and precise processing and analysis of software requirements. Its major advantages are as follows.

- 1. Custom Requirements:** So SUTRAVA uses a BERT-based Named Entity Recognition (NER) model, which we have specially fine-tuned for our Software Requirement Engineering tasks. This custom model after the fine-tune can now understand requirement related entities, such as stakeholders, architectural elements and both functional and non-functional requirements.
- 2. Semantic Clustering:** Same AI also helps us to cluster different requirements into different clusters like functional, mix and non-functional requirements. This happens as the model finds relationships between words and makes them a cluster.
- 3. Four-Layer Prioritization Framework:** SUTRAVA follows a four-layer prioritization framework that systematically sees the requirement urgency, validates semantic consistency, resolves conflict between stake-holders requests using Decision Tree, and generates an ambiguity score.

4. **End-To-End Traceability and Explainability:** Every generated Requirement directly connected to their respective stake-holders, this ensures the traceability of the requirements. This also explains how each requirement is generated.

4. How does it differ from existing solutions?

Although existing RE tools and LLMs can help in requirements generation or extraction, but they have some limitations when applied to automated requirements generations.

1. **Domain Specific Framework Vs Generic LLMs:** Generic LLMs are especially designed for Large Language Tasks and may generate inconsistent results, unseeing some critical information, or can produce some wrong prioritization decisions when handling large and complex collections of stakeholders Messages and Emails. SUTRAVA handles it very easily as it have a dedicated RE* architecture that applies deterministic processing method especially for the software engineering domain. And every generated Requirement is directly connected to their respective stakeholders, this ensures the traceability of the requirements. This also explains how each requirement is generated.
2. **Comprehensive End-To-End Automation:** A major unique point of our SUTRAVA is its ability to automate the complete process from the stakeholder message, tickets and mails to the Requirement Engineering document rather than focusing on isolated activities like copying the emails and messages to any AI and getting some incomplete results. As our platform uses **Four-Layer Prioritization Framework**, this helps our platform to get to the crux of the messages and email and get the correct result for the user. So by integrating these capabilities into one cohesive pipeline, SUTRAVA can get things from given information that is not given by any other platform as of now, like structured data from unstructured stakeholder messages, prioritisation of different requirements and explain Agile user stories.

Capability	Visure Solutions	Aqua Cloud	ReqSpell	SUTRAVA
NLP Requirement Extraction	Yes	Partial	Yes	Yes
FR/NFR Classification	Yes	No	No	Yes
Custom NER Entities	No	No	No	Yes
Semantic Clustering	No	No	No	Yes
Multi-Layer Prioritization	No	Basic	No	Yes
Explainability (XAI)	No	No	No	Yes

Requirement Summarization	No	No	No	Yes
Live Jira/Slack Integration	No	Limited	No	Yes
On-Premise Deployment	No	No	No	Yes
End-to-End RE Pipeline	No	No	No	Yes

Key Differentiators of SUTRAVA

Unique Capability	SUTRAVA
Custom Requirements NER Schema	Yes
Semantic Requirement Clustering	Yes
Explainable AI Decisions (XAI)	Yes
Multi-Layer Priority Intelligence	Yes
Abstractive Requirement Summarization	Yes
Live Communication Mining (Jira, Slack, Email)	Yes
Complete End-to-End Automation Pipeline	Yes
Self-Hosted / On-Premise Deployment	Yes

5. Research Prototype?

The SUTRAVA system was built as a research prototype to demonstrate the feasibility of using the presented framework. With the help of a dedicated dashboard, the system consumes the unstructured information of stakeholders, such as the exported data from Jira, processes the information through all eight phases of the pipeline and outputs structured software requirements, sorted by importance, along with confidence levels for every requirement.

Though the framework is based on research, the main emphasis in it lies in practice. Thus, SUTRAVA is especially

useful for small and medium enterprises (SMEs), as well as for startups, that often do not have Business Analysts and Requirements Engineers. Automating some requirements engineering functions allows reducing the amount of time and efforts spent for transforming the information provided by stakeholders into structured software requirements. Moreover, SUTRAVA outputs the information in structured JSON format that can be imported into Jira and Trello.

6. Stakeholders and Utilization

Small and Medium Enterprises and Startups: They can use SUTRAVA for automating requirement engineering and also ease the workload on their small engineering teams hence preventing the expensive miscommunications.

- **Product Owners and Project Managers:** They can use SUTRAVA to look at a lot of feedback from stakeholders. This helps because it does the work for them. They do not have to do it manually which saves time and effort hence reducing the overall cost. SUTRAVA marks the important requirements so they are not missed and thought about.
- **Business Analysts:** They can use this to look at requirements that do not agree with each other. They can fix these problems. Make sure everything is correct. They check to see if they can trace things back and understand them. Then they send the requirements to the people who will develop the software.
- **Software Developers and Architects:** They receive well-structured and unambiguous user requirements/stories with tagged system constraints and technical entities, making it easier to understand and implement the requirements during the software development.

7. References

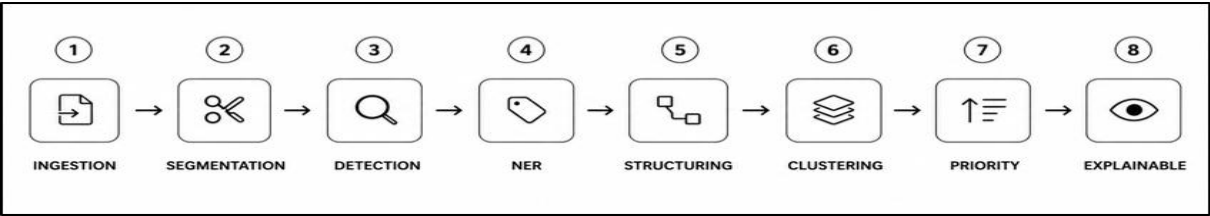
The development of SUTRAVA is based on existing research in AI-driven Requirements Engineering (RE), but it also introduces several new ideas. The proposed framework addresses some of the limitations identified in earlier studies by applying domain-specific AI techniques throughout the Requirements Engineering process. Its design is inspired by previous research on intelligent requirements elicitation and analysis, which has guided the development of the framework.

[1] **Transformer-Based Models for Named Entity Recognition, 2023. (Demonstrates foundational NER approaches that SUTRAVA adapts specifically for RE).**
[2] **Automated Classification and Identification of Non-Functional Requirements in Agile-Based Requirements Using Pre-Trained Language Models, 2023.**
[3] **NLP4RE: Natural Language Processing for Requirements Engineering, 2020.**
[4] **R. Ali et al., A Goal-Based Framework for Contextual Requirements Modeling and Analysis.**
[5] **A Systematic Review of AI-Enabled Frameworks in Requirements Elicitation, 2024.**
[6] **R. Ali et al., RECOVER: Toward Requirements Generation from Stakeholders' Conversations, 2024.**

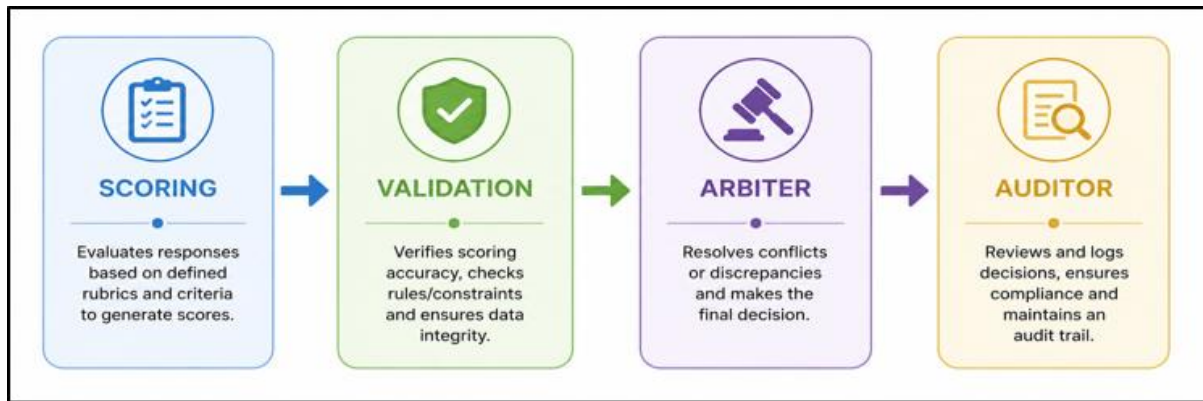
8. System Architecture & Visual Proofs

8.1 System Architecture Diagram

The SUTRAVA framework's whole workflow is shown in the following picture, which shows how unstructured stakeholder inputs are converted into structured, prioritized software requirements via an eight-stage AI-driven processing pipeline. The finished product is produced in JSON format, which offers a machine-readable, standardized representation that is easily incorporated into software development processes.



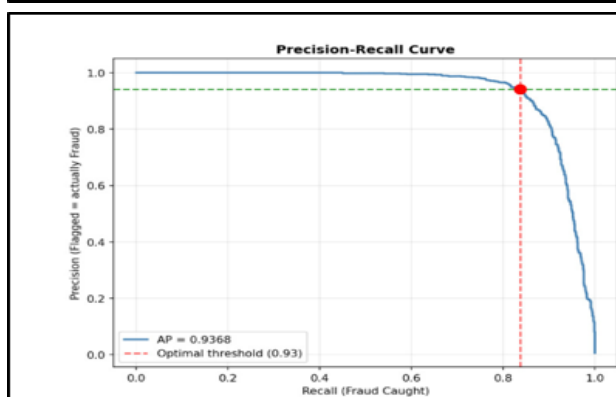
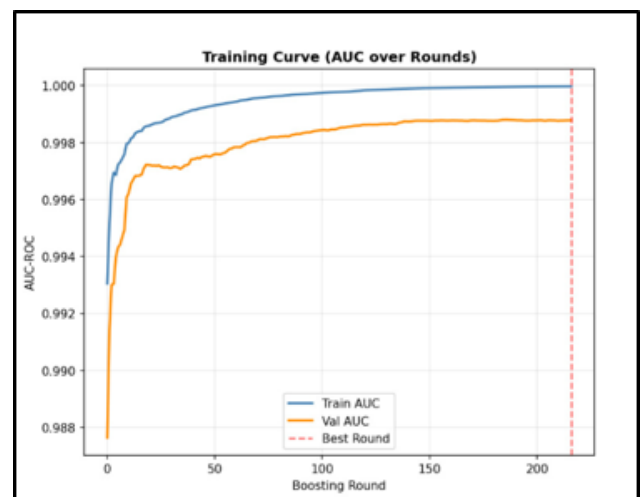
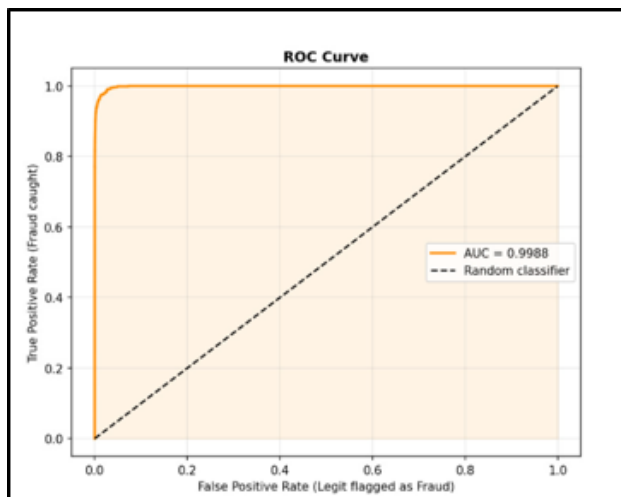
8.2 The 4-Layer Prioritization Flowchart



The SUTRAVA decision-making procedure involves four steps which include the Scoring module, the Validation module, the Decision Arbiter module, and the LLM Auditor module, as shown in the flow diagram below. The whole procedure aims to evaluate the stakeholder requirements, ensure consistency of the requirements, resolve any conflicting requests and set priority level. Confidence scores and ambiguity measures have been incorporated into the framework to increase the transparency and explainability of the requirements prioritization process.

8.3 Machine Learning Performance Metrics

Below are the graphs showing the performance evaluation of SUTRAVA's Requirement Named Entity Recognition (NER) model. They show how effectively the model can recognize software engineering entities from unstructured stakeholder inputs using important performance measures like Precision and Recall. The graphs prove that the proposed model is effective in extracting domain-specific information and requirements engineering.



8.4 Research Prototype Functional UI

The following screenshots present an example of how SUTRAVA has been successfully implemented as a fully-functional research prototype. It can be seen how SUTRAVA is capable of clustering requirements based on their semantic relation, measuring the ambiguity of requirements, and generating explainable priority values. The capabilities presented above prove the usefulness of SUTRAVA in Requirements Engineering automation.

